



◀ Sometimes, the designers went a wee bit overboard

marking the time when the space race between the USA and the USSR started ramping up, Alan Shepard became the second man in space aboard the Mercury Redstone 3. He wore the Navy Mark IV, a suit modified by NASA from the US Navy's pilots of High Altitude Aircraft. It weighed 22 pounds but it was still too bulky and uncomfortable to move in, and had bad temperature regulation. The first EVA suit (all the others before this were IVA suits) was the Berkut, worn by the astronauts aboard the Vokshod 2 spacecraft. It was a modified SK-1 with a 'backpack' containing about 45 minutes of oxygen for use on the first ever space walk. Berkut was used for only one mission since moving in it was really difficult and created a problem when the astronaut couldn't get back into the airlock. Then technology improved rapidly and features such as liquid cooling, better insulation, fire-proofing (after the disastrous Apollo-1 fire) started making their way into development. The A7L Apollo & Skylab spacesuit was the culmination of many years of developing and was claimed as a pinnacle of engineering. It was used in the first lunar landing and featured astonishingly advanced hardware for the time including a *Liquid Cooling and Ventilation* loop for thermal regulation, synthetic and rubber joints to allow

unparalleled mobility relative to other spacesuits and the now famous "fish-bowl" helmet, which allowed for nearly unobstructed vision. Development went on at a furious rate for the next years before peaking in the late 70's and early 80's. Suits developed during that time – Extravehicular Mobility Unit from the US and the Orlan Spacesuit from Russia are still being used today with almost no major changes to the design.

WHAT IS IT MADE OF?

Space suits are made up of several different, distinct pieces. The helmet is made of a strengthened polycarbonate which protects the astronaut as much as it can while not impeding vision. The visor is further coated with a thin layer of gold to act as UV protection and there is a neck ring and a ventilation system. The helmet also has a straw to a drinking bag. Some astronauts stick a piece of velcro with the rough side up inside their helmets for when they get a nose itch in space to prevent them from going batshit insane (or so they say). Then there is the 'Snoopy Cap' or Communications Carrier Assembly which is basically a covering worn on the head that houses the microphone and radio. The camera and lights which are usually mounted on the helmet also link to this. The Upper Hard Torso which

is made of fiberglass provides rigid support for all the other components and mounting for the extremely vital 'backpack'. This backpack or the *Primary Life Support Subsystem* is the housing for several major components. There are two oxygen tanks – one for regular use which lasts upto seven hours and one for emergency which holds upto 30 minutes worth of air. You'll also find CO2 filters, a water reservoir to regulate the temperature in the whole body and the battery pack. It also houses the *Simplified Aid for Extravehicular Activity* or SAFER device which is essentially a series of mini-thrusters an astronaut can use in a spacewalk to navigate incase the umbilical cord tethering them to the spacecraft snaps off somehow. The *Arms Assembly* and *Lower Torso Assembly* protect the arms and legs respectively with the lower torso assembly taking care of the urine collection system as well (Hey, don't judge, a space walk can take up to seven hours!). There are actually several layers inside a spacesuit. The lower most layer is the *Maximum Absorption Garment* or MAG which is essentially an adult sized diaper to collect excreta. Over it an underlayer made out of spandex, an incredibly stretchy material, covers the whole body. The 'bladder' or the layer which traps the gas inside the suit is usually made of urethane-coated nylon. A restraining layer to make the bladder conform to a shape is made up of Darcon and then the outer thermal layer is composed of Neoprene-coated nylon. Nearly 300 feet of plastic tubing is interwoven between the layers. This is used as a liquid cooling solution and it is maintained at around 10 degrees Celsius. It also has anywhere from five to eight layers of aluminised Mylar and a fabric surface layer composed of Teflon, Kevlar, and Nomex. The outermost layer is reflective to prevent overheating from the Sun's rays and protect the astronaut from radiation. Over 80 companies produce the parts for one spacesuit and sets back the space agency a cool 12,000,000 USD.